

Spring 2024 Exam

ST443

Machine Learning and Data Mining

Suitable for all ST candidates

Instructions to candidates

This paper contains five questions. Answer **ALL Five**.

Question 1: 21 marks

Question 2: 18 marks

Question 3: 21 marks

Question 4: 22 marks

Question 5: 18 marks

The marks in brackets reflect marks for each part of a question.

Candidates must not take the question paper away with them after the examination. Please leave it on your desk at the end of the examination.

Time allowed **Reading Time:** *None*
 Writing Time: *2 hours*

You are supplied with: No additional materials

You may also use: No additional materials

Calculators: *Calculators are allowed in this exam*

1. (a) Select *True* or *False* for the following statements. Briefly justify your answers.
- i. Support vector machines always have zero training error because they will find a separating hyperplane in the feature space. [3 marks]
 - ii. A natural cubic spline with K knots (including both internal and boundary knots) has K degrees of freedom. [3 marks]
 - iii. The mean squared error of an estimator is equal to the bias of the estimator plus the variance of the estimator. [3 marks]
- (b) Give an example of the Bootstrap aggregation (Bagging) approach for constructing classification trees such that using "majority vote" and "taking the average of probabilities" will give different classification results. [6 marks]
- (c) Let $\mathbf{x} = (x_1, x_2)^T$ and $\mathbf{y} = (y_1, y_2)^T$. Show that the polynomial kernel of degree 2 defined as $K(\mathbf{x}, \mathbf{y}) = (1 + x_1 y_1 + x_2 y_2)^2$ is positive definite symmetric. [6 marks]

2. Consider the multiple linear regression without the intercept term

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{Y} = (y_1, \dots, y_n)^T$, $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$, $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^T \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ and $\mathbf{X} = (x_{ij})_{n \times p} \in \mathbb{R}^{n \times p}$ is the design matrix. The ridge regression estimator solves the following minimization problem:

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_{j=1}^p \beta_j^2,$$

where $\lambda > 0$ is the regularization parameter. In class, we have shown that the solution is

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^T \mathbf{Y}.$$

- (a) Compute the bias and variance of the ridge estimator $\hat{\boldsymbol{\beta}}_{\text{ridge}}$. [8 marks]
- (b) Let $\hat{\mathbf{Y}} = (\hat{y}_1, \dots, \hat{y}_n)^T$ be the fitted value for $\mathbf{Y}$. Find $\mathbf{S} \in \mathbb{R}^{n \times n}$ such that $\hat{\mathbf{Y}} = \mathbf{S}\mathbf{Y}$. [2 marks]
- (c) The Generalized Cross Validation (GCV) provides an approximation to Leave-One-Out Cross Validation Error, the minimization of which can be used to select the optimal regularization parameter $\hat{\lambda}$. Note that

$$\text{GCV}(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - \text{trace}(\mathbf{S})/n} \right)^2.$$

Let $d_1 \geq \dots \geq d_p > 0$ be the eigenvalues of $\mathbf{X}^T \mathbf{X}$. Express $\text{trace}(\mathbf{S})$ as a function of $d_1, \dots, d_p$ and λ . [8 marks]

3. (a) The table below is the training data with $n = 4$ observations of an numerical input X and a binary output Y (the color is blue or red).

i	X	Y
1	0	red
2	-2	red
3	3	blue
4	1	blue

Find the Leave-One-Out Cross Validation error when using k -nearest neighbours for $k = 1, 3, 4$. [8 marks]

- (b) Using the data below, we want to build a classification tree which classifies Y as True or False (T or F) given the binary variables A, B, C (each with value T or F).

A	B	C	Y
F	F	F	F
T	F	T	T
T	T	F	T
T	T	T	F

- i. Draw the classification tree that is learned by recursive binary splitting algorithm with zero misclassification error. You do not need to show any computation. [7 marks]
 - ii. Draw another differently structured classification tree with zero misclassification error. You do not need to show any computation. [6 marks]
4. Consider a three-class classification problem with response $Y = 1, 2, 3$ respectively corresponding to classes 'Red', 'Blue', 'Orange' with equal probabilities and observation $\mathbf{X} \in \mathbb{R}^2$ with conditional distribution

$$\begin{aligned} \mathbf{X}|Y = 1 &\sim N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}), \\ \mathbf{X}|Y = 2 &\sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}), \\ \mathbf{X}|Y = 3 &\sim N(\boldsymbol{\mu}_3, \boldsymbol{\Sigma}), \end{aligned}$$

where $\boldsymbol{\mu}_1 = (0, 0)^T$, $\boldsymbol{\mu}_2 = (1, 1)^T$, $\boldsymbol{\mu}_3 = (-1, 1)^T$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

- (a) What is the posterior probability of the point $x_0 = (0, 1)^T$ belonging to each of the three classes? [6 marks]
- (b) Describe the Bayes classification boundary in this classification problem. [6 marks]
- (c) Suppose now that observations with $Y = 2$ ('Blue' class) and $Y = 3$ ('Orange' class) are grouped into a single class 'BlueOrange' corresponding to $Y = 2$ or 3, so that data are viewed as generated from a two-class classification problem and

$$\begin{aligned} \mathbf{X}|Y = 1 &\sim N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}), \\ \mathbf{X}|Y = 2 \text{ or } 3 &\sim \frac{1}{2}N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}) + \frac{1}{2}N(\boldsymbol{\mu}_3, \boldsymbol{\Sigma}), \end{aligned}$$

where the conditional distribution of $\mathbf{X}$ given $Y = 2$ or 3 is a mixture of two Gaussians. Classify the point $x_0 = (0, 1)^T$ in this two-class problem based on the posterior probabilities. [6 marks]

- (d) Have the Bayes classification region for the 'Red' class (i.e., the set of x values that would be classified as 'Red' using posterior probabilities) changed when we moved from the three-class problem to the two-class problem in part (c)? Please explain. [4 marks]
5. (a) Suppose that we have five observed points, each with three features. We present the Euclidean distance between any two observations with measurements on these three features in the following matrix.

	1	2	3	4	5
1	0	1.9	2.5	1.7	2.6
2	1.9	0	0.8	2.8	2.0
3	2.5	0.8	0	3.5	1.9
4	1.7	2.8	3.5	0	2.9
5	2.6	2.0	1.9	2.9	0

Use the matrix with Euclidean distances to perform hierarchical clustering. Draw the corresponding Dendrogram using complete linkage. [10 marks]

(b) Suppose that we are given the following dataset

$$\{(0, 4), (1, 3), (2, 6), (3, 3), (5, 5), (5, 6), (9, 9)\}.$$

We apply k -means, with $k = 3$, to cluster the dataset using the Manhattan distance for computing the distances between centroids and pairs in the dataset. The Manhattan distance between (x_1, y_1) and (x_2, y_2) is defined as

$$|x_1 - x_2| + |y_1 - y_2|.$$

Suppose that the initial clusters of k -means are C_1 , C_2 and C_3 with

- $\{(1, 3), (3, 3), (5, 6)\},$
- $\{(0, 4), (2, 6)\},$
- $\{(5, 5), (9, 9)\},$

and that the centroids for C_1 , C_2 and C_3 are $(3, 4)$, $(1, 5)$ and $(7, 7)$ respectively. If we then run k -means for two iterations, what are the new clusters and what are their centroids? [8 marks]